Wetland Ecological Condition & Change Assessment

Spatial assessment of wetland vegetation condition, continuous spectral dynamics, and surface-water evidence across the configured observation interval.

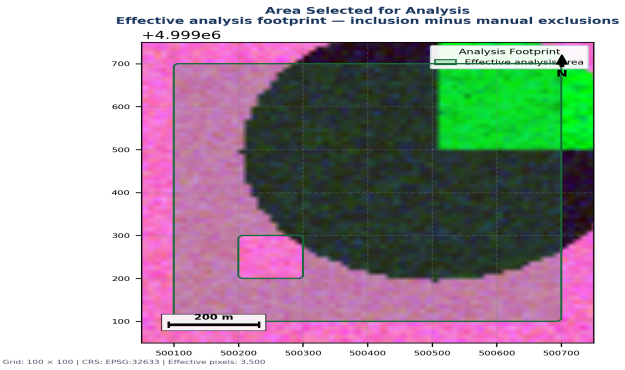
OBSERVATION WINDOW	2023-08-14 → 2023-08-28	INTERVAL	14 days
WETLAND CONTEXT	Project-provided wetland boundary	RUN ID	run_20260922_082400
ANALYSIS BASIS	RASTER_CRS_AND_TRANSFORM	SPATIAL REFERENCE	EPSG:32633

KEY FINDINGS

Measurable Ecological Change Detected in Analysis Footprint

The assessment identifies measurable vegetation change within the analysis area (0 pixels of degradation, 0 pixels of recovery). These shifts are evaluated alongside spectral and hydrological indicators below. Surface-water dynamics indicate 0.00 ha of contraction (drying) and 3.78 ha of expansion (wetting). Physical wetland loss is not established. In accordance with rigorous scientific standards, optical vegetation change alone is not interpreted as permanent physical wetland loss.

INDICATOR	OBSERVED VALUE
Vegetation Degradation	0 px · 0.00%
Vegetation Recovery	0 px · 0.00%
Stable Classified State	3,500 px · 100.00%
CVA Magnitude (Mean)	0.6969
Effective Footprint	3,500 px



Analysis Footprint / AOI

1. THE AREA WE ANALYZED

This assessment evaluates ecological condition within the boundary selected for analysis. Changes in vegetation canopy, condition, and surface water are evaluated between the two observation dates. The analysis measures condition changes within this defined spatial footprint; it does not constitute a statutory wetland delineation or field boundary survey.

The analysis is spatially bounded by the project-configured wetland boundary. All ecological condition indicators are evaluated within this defined area.

SPATIAL ATTRIBUTE	OBSERVED / CONFIGURED VALUE
Wetland context status	WETLAND_BOUNDARY_AVAILABLE
Wetland source	Project-provided wetland boundary
Source dataset / boundary	wetland_boundary.geojson
Total analysis area	36.00 ha
Effective analysis footprint	35.00 ha
Physical depletion pathway	NOT_ASSESSED

SOURCE IMAGERY (CIR FALSE-COLOUR COMPOSITES)

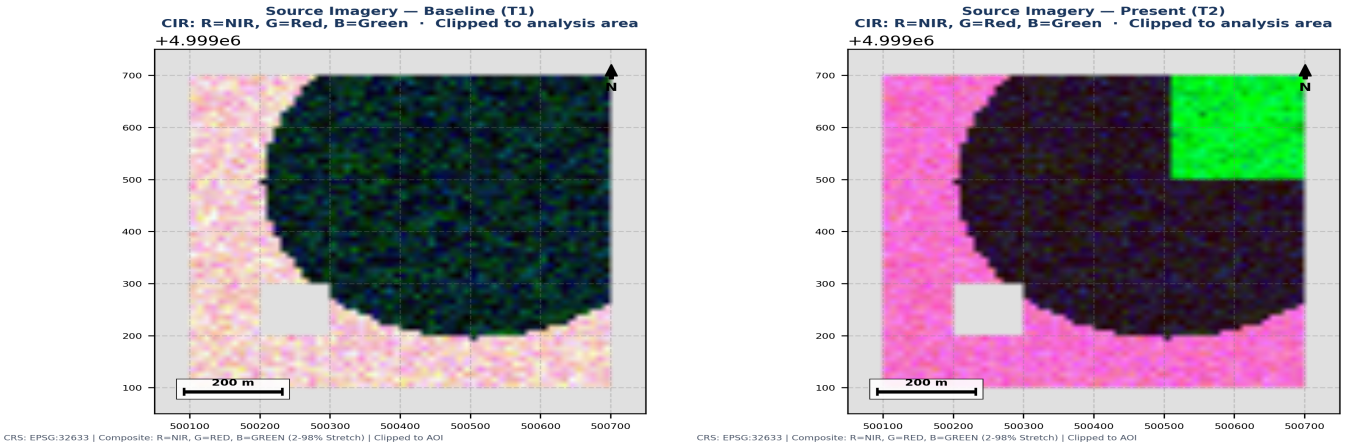


Figure 1: Baseline (T1, 2023-08-14) and present (T2, 2023-08-28) multispectral false-colour composites (NIR-Red-Green), clipped to the analysis footprint.

ANALYSIS FOOTPRINT AUDIT

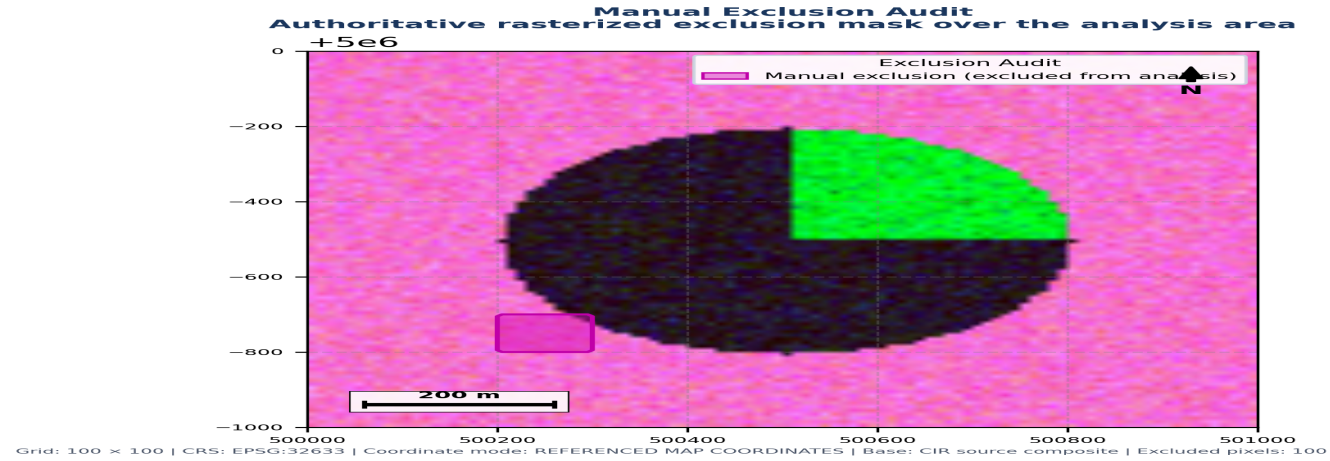


Figure 2: Authoritative manual exclusion audit map showing removed areas (100 pixels).

2. VEGETATION CONDITION

Vegetation condition within the analysis area is classified at baseline (T1) and present (T2) into four condition classes using the multispectral ResNet34 U-Net model: Healthy Vegetation, Moderate Stress, Severe Stress, and Background / Non-Vegetated.

CONDITION CLASS	T1 PIXELS	T1 %	T2 PIXELS	T2 %	CHANGE
Healthy vegetation	0	0.00%	0	0.00%	0 px
Moderate stress	0	0.00%	0	0.00%	0 px
Severe stress	0	0.00%	0	0.00%	0 px
Background / non-vegetated	10,000	100.00%	10,000	100.00%	0 px

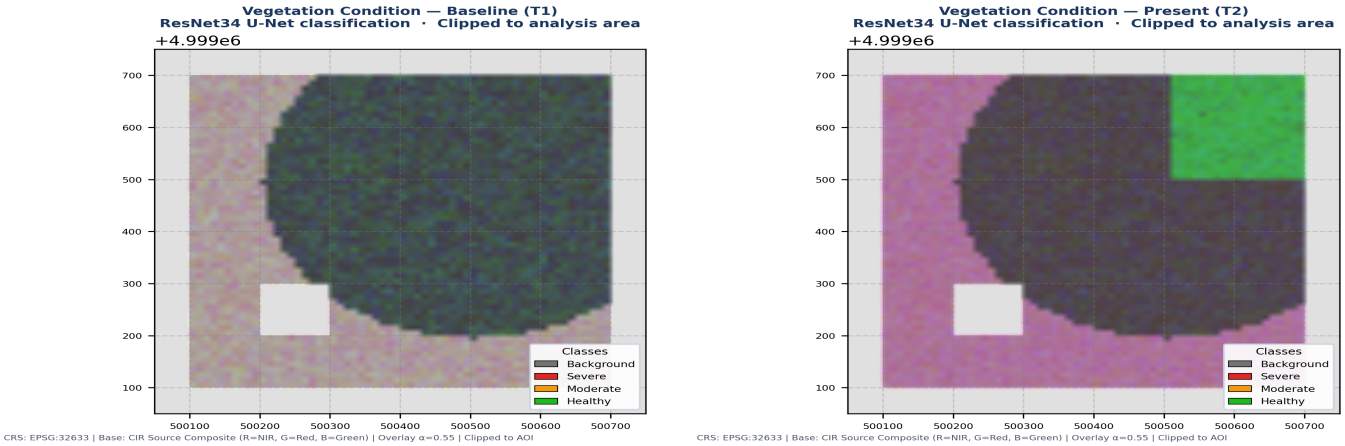


Figure 3: Classified vegetation condition at baseline (T1, 2023-08-14, left) and present (T2, 2023-08-28, right).

Vegetation condition summary: Aggregate classified vegetation states remained constant across the interval (Healthy Vegetation: 0 pixels). No pixels entered degradation transition states.

3. VEGETATION CHANGE & TRANSITIONS

Pixel-by-pixel transition tracking identifies the exact trajectory of vegetation change between observation dates, distinguishing defined degradation pathways (movement toward higher stress) from recovery, canopy loss, or persistent stable canopy.

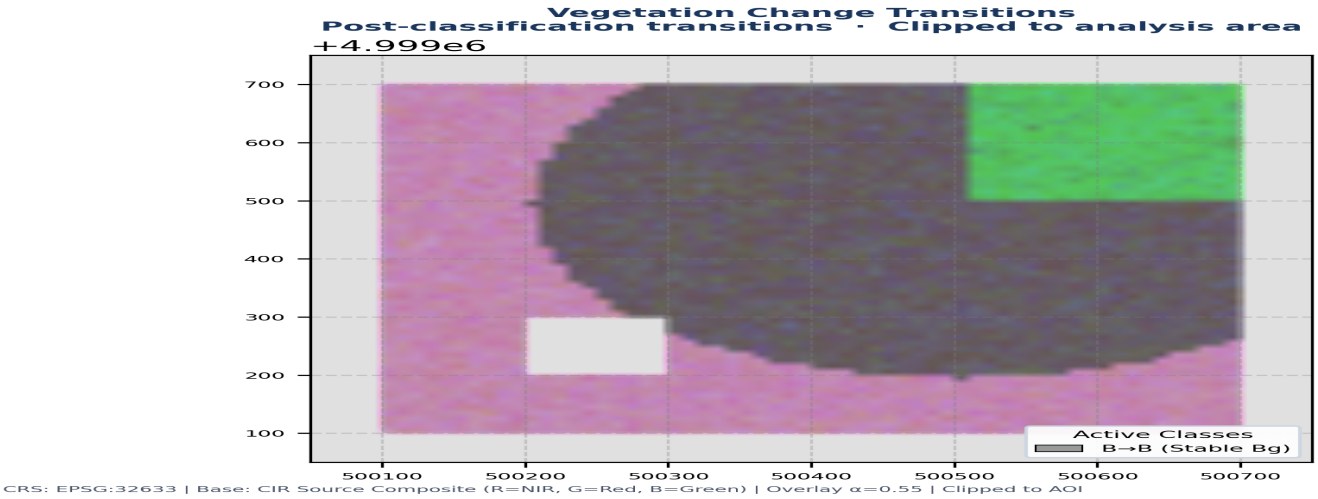


Figure 4: Post-classification vegetation transitions across the analysis area.

TRANSITION PATHWAY	DEFINED TRANSITIONS	PIXELS	PROPORTION
Persistent classified state	H→H, M→M, S→S, B→B	3,500	100.00%
Movement toward poorer vegetation condition	H→M, H→S, M→S	0	0.00%
Movement toward better vegetation condition	M→H, S→M, S→H	0	0.00%
Canopy loss / non-vegetated transition	H→B, M→B, S→B	0	0.00%
Canopy emergence / colonisation	B→H, B→M, B→S	0	0.00%

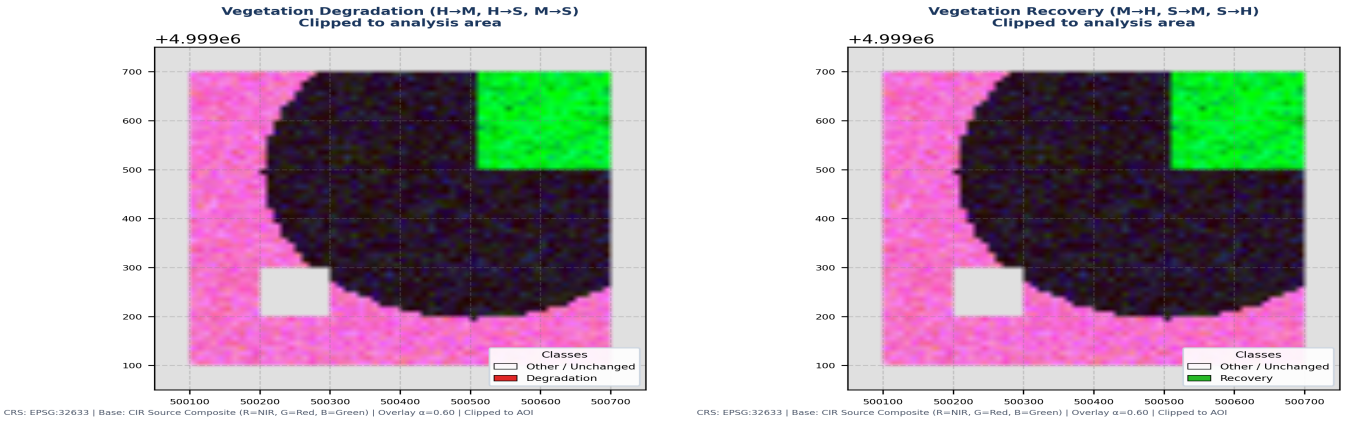


Figure 5: Spatial distribution of detected vegetation degradation (left) and recovery (right).

Transition summary: Zero pixels followed defined degradation pathways (H→M, H→S, M→S), and zero pixels followed recovery pathways. The classified vegetation landscape remained temporally stable.

4. CONTINUOUS SPECTRAL CHANGE

Continuous spectral indicators measure radiometric shifts across the multispectral feature space. Change Vector Analysis (CVA) integrates ΔNDVI, ΔGNDVI, and ΔNDRE to detect physiological canopy changes that may occur within or across classification boundaries.

SPECTRAL INDICATOR	MEAN	MEDIAN	P95
CVA Magnitude	0.6969	0.6719	1.3386
ΔNDVI	0.4944	0.2951	1.0850
ΔGNDVI	0.2689	0.0051	0.8085
ΔNDRE	-0.0334	-0.0046	0.3494

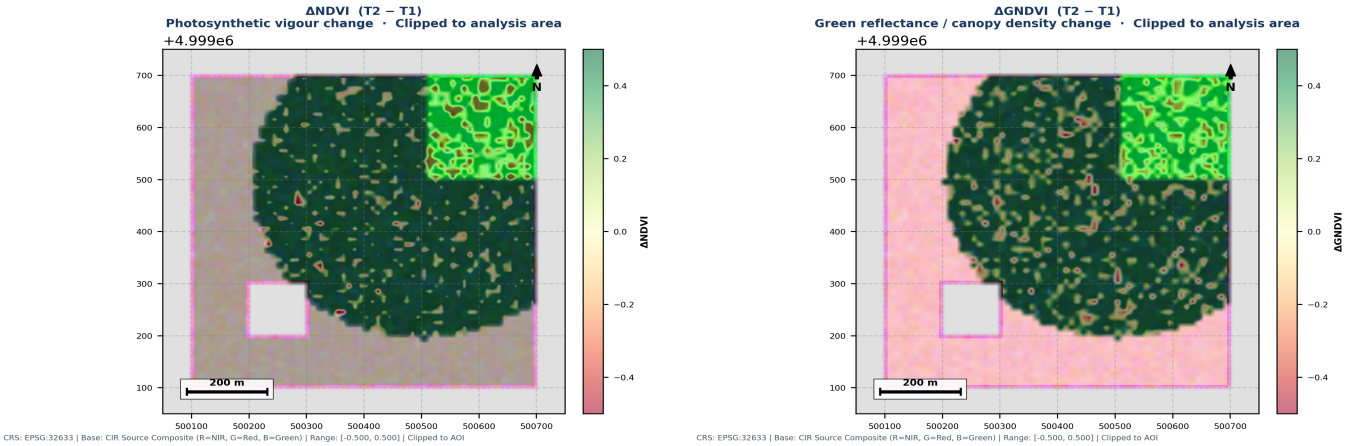


Figure 6: Continuous vegetation index change (ΔNDVI, left; ΔGNDVI, right) between T1 and T2.

Spectral summary: Continuous spectral indicators record a mean CVA magnitude of 0.6969. This reflects sub-classification physiological variation across the canopy.

5. SURFACE WATER DYNAMICS

Surface water and moisture dynamics provide an independent hydro-ecological line of evidence. Water presence is modelled from the Green/NIR Normalized Difference Water Index (NDWI) using a calibrated threshold of 0.00.

HYDROLOGICAL METRIC	OBSERVED VALUE	HYDROLOGICAL STATE DEFINITION
Baseline Water Extent (T1)	0.27 ha	Water classified above NDWI threshold at baseline
Present Water Extent (T2)	4.05 ha	Water classified above NDWI threshold at present
Surface Water Expansion (Gain)	3.78 ha	Newly detected surface water / wetting
Surface Water Contraction (Loss)	0.00 ha	Previously detected water becoming non-water / drying
NDWI Water Threshold	0.00	Standard McFeeters zero-plane calibrated threshold

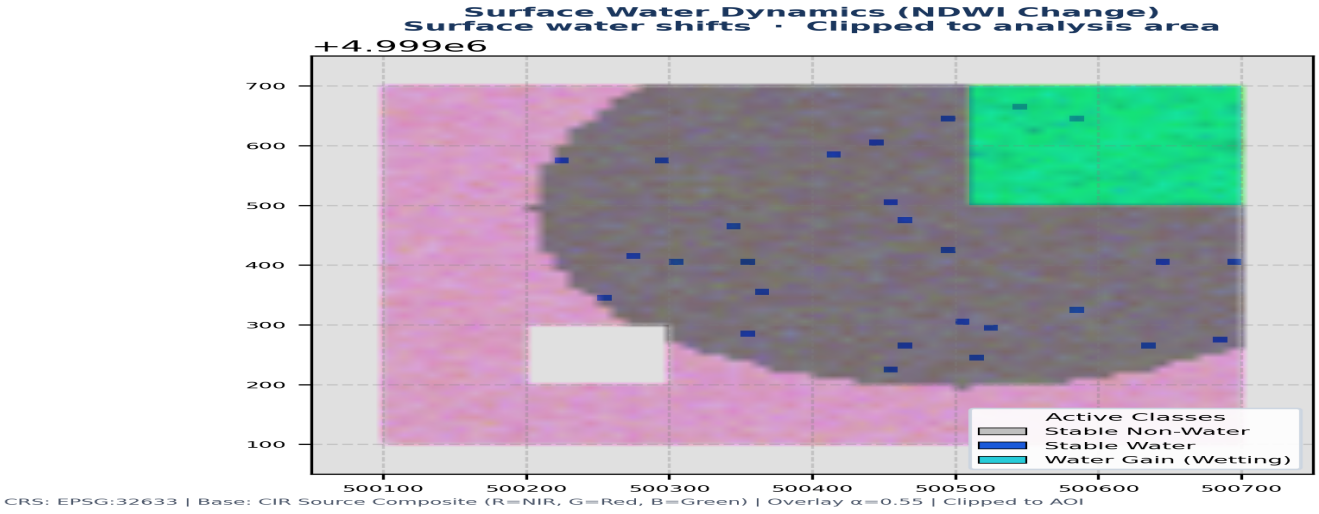


Figure 7: Surface water dynamics (wetting, drying, and stable water) across the analysis area.

Hydrological summary: Surface-water dynamics indicate 0.00 ha of contraction (drying) and 3.78 ha of expansion (wetting).

6. INTEGRATED ASSESSMENT & SYNTHESIS

The findings below synthesize all independent lines of evidence evaluated in this run: vegetation condition transitions, continuous spectral change, and surface-water dynamics.

EVIDENCE STREAM	OBSERVATION	SUPPORTED INFERENCE
Wetland context	Project-provided wetland boundary	Establishes the spatial boundary within which ecological observations are evaluated.
Vegetation transitions	0 degradation px; 0 recovery px	Indicates whether classified vegetation condition shifted toward stress or recovery.
Continuous spectral change	CVA mean 0.6969	Indicates whether average canopy spectral reflectance shifted within classes.
Hydrological evidence	Gain 3.78 ha; Loss 0.00 ha	Indicates whether surface moisture and water presence corroborate vegetation changes.

The T1–T2 comparison identifies measurable vegetation change within the analysis area. These shifts represent observable vegetation dynamics that must be interpreted alongside hydrological evidence to distinguish transient variation from sustained degradation.

Continuous spectral indicators (mean ΔNDVI: 0.4944, ΔGNDVI: 0.2689, ΔNDRE: -0.0334) provide fine-scale evidence of canopy and foliar condition shifts.

Physical wetland loss is not established. In accordance with rigorous scientific standards, optical vegetation change alone is not interpreted as permanent physical wetland loss.

CONCLUSION: MEASURABLE VEGETATION CHANGE DETECTED

This conclusion describes detected condition changes within the specific analysis boundary. It reflects remote-sensing evidence over the observation window and does not constitute a regulatory determination of wetland status or permanent physical extent.

7. MULTI-EVIDENCE FUSION

Evidence fusion cross-references vegetation transitions with surface-water shifts. A composite hydro-ecological degradation signal requires concurrent evidence of both vegetation stress and surface-water loss, preventing single-sensor optical artifacts from triggering false alarms.

EVIDENCE CLASS	EXTENT / COUNT	DEFINITION & MEANING
Stable landscape	3,500 px	Persistent classified condition state across both dates
Hydro-ecological degradation	0.00 ha	Concurrent vegetation degradation and surface-water loss
Hydro-ecological recovery	0.00 ha	Concurrent vegetation recovery and surface-water gain
Vegetation-only degradation	0 px	Vegetation condition decline without concurrent hydrological loss

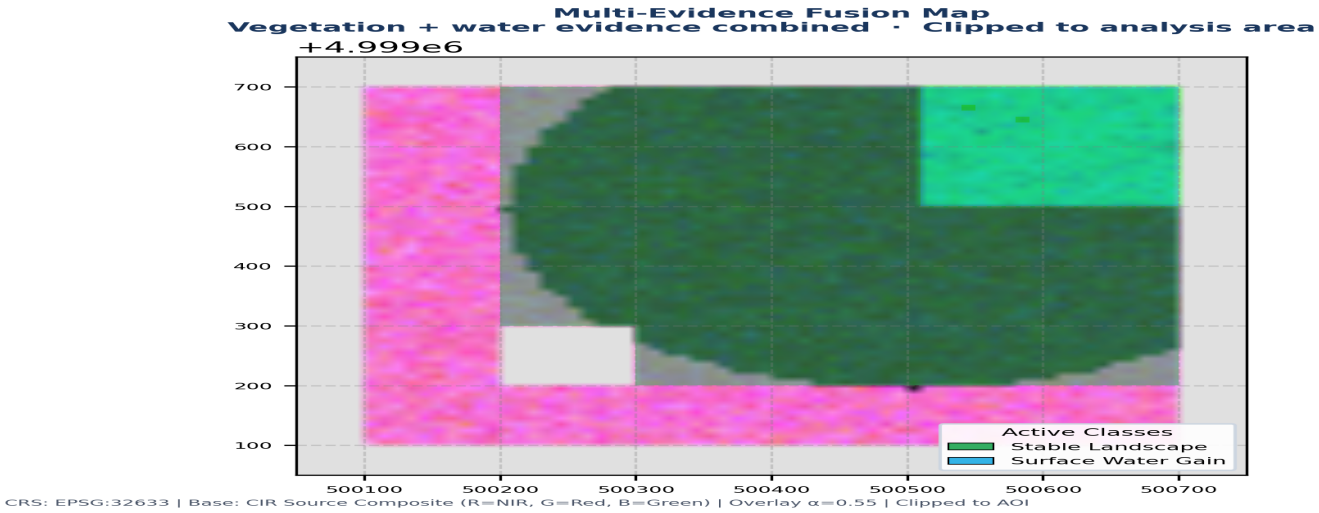


Figure 8: Multi-evidence fusion combining vegetation transitions and surface-water dynamics.

8. DATA QUALITY & SPATIAL ACCOUNTING

Spatial accounting verifies that every pixel is traced from the source imagery through the inclusion boundary and manual exclusions to the effective analysis footprint.

QUALITY CONTROL CHECK	STATUS	FINDING
Spatial grid alignment	PASS	T1 and T2 rasters were verified on a common analysis grid without spatial distortion.
Footprint reconciliation	PASS	10,000 px overlap = 100 px excluded + 3,500 px effective.
Independent ground reference	UNAVAILABLE	No external ground-truth validation plots were supplied for this run.
Physical depletion guard	ENFORCED	Optical vegetation decline is prevented from being reported as physical wetland loss.

SPATIAL FOOTPRINT RECONCILIATION

FOOTPRINT COMPONENT	PIXELS	PHYSICAL AREA
Original spatial overlap	10,000	100.00 ha
Selected AOI / Inclusion	3,600	36.00 ha
Manual exclusion	100	1.00 ha
Effective analysis footprint	3,500	35.00 ha

9. TECHNICAL PROVENANCE & LIMITATIONS

Every metric reported in this document is derived directly from verifiable raster artifacts produced during the processing pipeline. No analytical values are interpolated or inferred without evidence.

INDICATOR	VALUE	UNIT	PIPELINE ARTIFACT / METHOD
Wetland source	PROJECT_BOUNDARY	class	Wetland mask source selected by wetland context stage
Wetland context status	WETLAND_BOUNDARY_AVAILABLE	status	Wetland mask result
Effective analysis footprint	3,500.00000	pixels	alignment/effective_analysis_mask.tif
Manual excluded	100.00000	pixels	exclusions/manual_exclusion_mask.tif
Vegetation degradation	0.00000	pixels	degradation/degradation_map.tif (H→M, H→S, M→S)
Vegetation recovery	0.00000	pixels	degradation/recovery_map.tif (M→H, S→M, S→H)
CVA Mean Magnitude	0.69691	index units	change/cva_magnitude.tif (||[ΔNDVI, ΔGNDVI, ΔNDRE]||)
Mean ΔNDVI	0.49444	index units	change/delta_ndvi.tif (NDVI_T2 – NDVI_T1)
NDWI Water Threshold	0.00000	index units	Green/NIR NDWI threshold
Physical depletion assessment	NOT_ASSESSED	status	Independent-evidence claim guard

SCIENTIFIC LIMITATIONS & BOUNDARIES

Observation Interval: Remote-sensing comparisons reflect conditions at specific acquisition times. Short-term intervals capture immediate physiological or seasonal responses that may differ from multi-year ecological trends.
Model Inferences: Machine learning vegetation classifications are probabilistic estimates. In-situ field validation is recommended before using findings for regulatory, legal, or compensation purposes.
Spatial Reference: Where imagery lacks a validated Coordinate Reference System (CRS), pixel counts serve as the authoritative spatial unit. Physical area (hectares) is reported only when georeferencing is confirmed.

Run: run_20260922_082400 · **Software:** GIS Engine v1.0.0 · **Analysis Basis:** RASTER_CRS_AND_TRANSFORM